

Marek ZAREMBA-PIKE, MBCS

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Staff software engineer building and deploying production AI systems at scale. Led high-throughput backend platforms (2,500+ RPS) and designed LLM integrations with safeguards including rate limiting, fallbacks, and abuse prevention. Currently focused on building and optimising agentic coding workflows.

Technologies

Python, Django, FastAPI

Kubernetes, Lambda

Managed projects in Angular, PHP, React, Vue.js

AWS, Google Cloud Platform, DigitalOcean &

private data centres/on-prem

Microservices, event-driven arch., data pipelines

Production applications with Claude, GPT, Gemini,

Black Forest Labs, and internal AI APIs

GitHub Actions, GitLab's CI/CD

SNS/SQS, custom message queues, AWS Step Functions

DynamoDB, ElasticSearch, PostgreSQL, Redis

Recent Work Experience

12/2024 –
present

Staff Engineer, Head of Backend – Potoroom (AI & ECommerce)

A hands-on (65% IC) role responsible for Potoroom's Django and FastAPI backend (2,500 requests per second, 10TB database) and managed 4 direct reports

Oversaw FastAPI AI gateway, handling frontend requests to internal and external AI endpoints, including model fallbacks, rate limiting, abuse prevention, enriching requests, and more to ensure safe and reliable model use at scale

Expanded our system monitoring (anomalies, SLOs, error rates) to improve incident detection and response times.

Led migration to Kubernetes, zero-downtime deployments (including data migrations), introduced automated deployments and a testing environment for client developers to use

Tech lead for cross-platform (Android, backend, iOS, and web) redesign of our billing system to closely align users' AI usage with the subscription fees

Built an agentic coding pipeline with structured evaluation, delivering 50-90% of merged PRs per week. 80%+ of PRs received <3 change requests by using context-specific skills, reference files, and deterministic checks. Pipeline monitored (incl. chain-of-thought) for optimisation.

08/2022 –
10/2024

Staff Engineer – Lantum (Healthcare & Finance)

Planned, led, and worked on the refactor of the entire backend (Python/Django), reducing complexity, upgrading libraries, hardening security, and enabling new features.

Oversaw tech support and incident response; trained the team, established reporting systems, and built better monitoring and logs. Resulted in reducing exposure of feature squads from 25% of their time to less than 1%, and reduced developer churn

Established and evangelised software processes, patterns, and ways of working

Designed and developed internal tools for operations to increase their efficiency

07/2021 –
07/2022

Technical Director (CTO) – TrackTrack (LegalTech)

Transformed the software architecture, infrastructure, team, and software development processes to allow rapid growth in clients. Some clients required on-prem deployments of our

product inside their DMZ.

Oversaw the breakup of the Python/Django software into serverless microservices in a multi-cloud environment, including on-prem client infrastructure.

Led projects to increase test coverage, introduce an automated deployment pipeline, and ensure complete documentation.

11/2019 -
06/2021

Consulting Software Engineer & Architect, European Public Affairs Technologies

Tech Lead in five successfully delivered projects. I guided clients on the best architecture for their needs, budgets, and maturity levels. I prepared documentation, wrote software, and performed code reviews. Notable projects included:

Redesigning and rewriting a legal tech start-up's previously prohibitively slow and inaccurate search engine, allowing them to launch.

Redesigning and rebuilding the data architecture and connections between a client's SAP, PIM, and e-commerce website, which allowed it to relaunch online sales

06/2016 -
12/2019

Backend Python/Django Engineer – European Public Affairs Technologies

Designed a serverless and multi-cloud (GCP, AWS, and DigitalOcean) architecture for data pipeline and UI for collating social (including Twitter's Firehose) and news media (Briefed_eu) for EU lobbyists

Developed the full project (backend & frontend), including a central Django project, several serverless apps, and a text analysis API delivered using Flask.

Example Personal Projects

11/2025 -
01/2026

Savin Hood Tax Calculator

Built a UK income tax calculator with a Rust and Python/Django backend, Typescript and Astro frontend entirely using Claude Code and Codex.

01/2015 -
06/2016

Co-Founder & Product Owner – European Public Affairs Technologies

Designed a content-sharing platform (thewonk_eu) for EU policy analysis, which reached 30% of the Total Achievable Market and was recognised by ComRes as a leading source of policy analysis for EU policymakers.

Education

04/2026

AI Safety Bootcamp, ML4Good

09/2024 – ...

MSc Computer Science, University of Bath

06/2024 – 09/2024

AI Safety Fundamentals Alignment Course, BlueDot Impact

03/2024 – 06/2024

AI Programming Nanodegree, Udacity

05/2021 - 10/2021

Engineering Leadership, Cornell University

2010 – 2011

MSc European Public Policy, University College London

2006 – 2010

B.A. Russian Studies & International Relations, University of Birmingham

Languages

English (Native) | Russian (Fluent) | French, Polish, and Spanish (Lapsed)